

Soil moisture variation and dynamics across a wildfire burn boundary in a loblolly pine (*Pinus taeda*) forest



M. Bayani Cardenas*, Michael R. Kanarek

Geological Sciences, The University of Texas at Austin, United States

ARTICLE INFO

Article history:

Received 11 April 2014

Received in revised form 8 July 2014

Accepted 11 July 2014

Available online 30 July 2014

This manuscript was handled by Corrado Corradini, Editor-in-Chief, with the assistance of Renduo Zhang, Associate Editor

Keywords:

Soil moisture

Wildfire

Electrical resistivity

Loblolly pine

Ecohydrology

Infiltration

SUMMARY

A year after the most destructive wildfire in Texas (USA) history which occurred in and around Bastrop State Park, we established a 165 m-long study transect, bridging burned and unburned areas, to study post-wildfire soil moisture dynamics. Soil moisture content (θ) was monitored indirectly approximately monthly for half a year using a variety of methods with different measurement scales including: 2D electrical resistivity (ER) imaging and surface and vertical profiles using probes which measure soil dielectric properties. The burned section, where the majority of loblolly pine trees were killed, had higher θ and lower ER whereas the unburned end which is still populated by live pine trees had lower θ and higher ER. This pattern persisted from the ground surface and down to ~2 m and through the study period even after a rainfall event which made the whole transect generally wetter but with the burned end showing a much stronger wetting response to the storm. The differences in θ cannot be explained by differences in soil texture with the burned end with sand soil and the unburned end with less permeable loamy sand. The differing results may be explained by loss of canopy cover and by reduced transpiration at the burned end where the dead roots may also potentially serve as macropores. Thus, after fires and until new vegetation cover has grown, the burned areas will store and transmit more water which could lead to increased groundwater recharge and promote the recovery or invasion of certain types of vegetation.

© 2014 Elsevier B.V. All rights reserved.

1. Introduction

Wildfires have profound local and global effects on ecological, geomorphic and hydrological processes and can lead to drastic economic losses and public health issues (Bowman et al., 2009). Thus, wildfire impacts will need to be further studied in light of observed and expected increasing occurrences (Pechony and Shindell, 2010; Westerling et al., 2006). Wildfires have a variety of direct and immediate impacts within the burned areas with many effects pertaining to soils and the soil hydrological regime (Bento-Goncalves et al., 2012; Certini, 2005; Gonzalez-Perez et al., 2004).

For example, soil water repellency is a well-known consequence of fire (DeBano, 2000; Doerr et al., 2000; Letey, 2001; Varela et al., 2010; Zavala et al., 2010). However, the effect is observed at varying degrees, and sometimes not at all depending on the fire conditions, especially the severity of the fire, the original organic matter content, and the texture of the soil (DeBano, 2000; Neary et al., 2005). The severity of water repellency in the surface soil decreases with time, and often does not substantially affect

infiltration beyond the first year after the fire (DeBano, 2000; Neary et al., 2005).

Other processes such as surface sealing and production of aggregates, along with water repellency, is responsible for observed reductions in infiltration rates (Doerr et al., 2000; Kinner and Moody, 2010; Neary et al., 2005; Robichaud, 2000). Reduction in infiltration rates is now an accepted consequence of fires, with even new mechanisms proposed such as reduction in hydraulic conductivity due to extremely dry conditions (Ebel and Moody, 2013). Despite the increasing consensus that fires affect infiltration rates (Ebel and Moody, 2013), this is an area where there are many questions remaining, particularly in terms of connecting the physics (i.e., moisture flow) to measurable properties and extrapolations beyond the plot scale (Moody et al., 2009). This is critical since a direct consequence of drastic changes to vegetation cover, and soil texture and water repellency due to fire is increased runoff (flash flooding) and erosion (Doerr et al., 2000; Moody and Martin, 2001a, b, 2009; Moody et al., 2013).

Fires can produce hydrologic and ecologic boundaries between severely burned and unburned or lightly burned areas. For example, the root zone in unburned areas have been shown to exhibit lower soil moisture content due to transpiration (Silva et al., 2006). In Australia, it was observed that a severely burned forest

* Corresponding author.

E-mail address: cardenas@jsg.utexas.edu (M.B. Cardenas).

had 41% lower evapotranspiration relative to an unburned forest 1–2 years postfire (Nolan et al., in press). Others have found that burned areas are wetter due mainly to reduced canopy interception and more throughfall (Soto and Diaz-Fierros, 1997). In shrublands in Spain, burned areas had 88% throughfall, while unburned areas had 58% throughfall (Soto and Diaz-Fierros, 1997); this led to higher soil moisture in the burned areas. Silva et al. (2006) similarly observed wetter soils in burned areas due to experimental fires. However, other systems displayed very minor differences in soil water content between burned and unburned areas when the soils tend to be generally dry (Obrist et al., 2004).

While the reduction or removal of root water uptake and increased rainfall (less interception) could counter the effects of water repellency (Silva et al., 2006), water repellent soils may still have large infiltration rates when macropores are present and potentially dominant (Nyman et al., 2014, 2010). Macropores are typically formed by dead and decaying roots (Beven and Germann, 2013; Noguchi et al., 1999), and one would expect these to be abundant in burned areas. Recent measurements showed increasing importance of macropore flow after fires (Nyman et al., 2014). Burned soils have also been found to promote more fingered and preferential flow (Stoof et al., 2014). Thus, fires lead to competing processes that control the soil water regime whose coupling and feedbacks require further study especially since wildfires have a major impact on soil moisture dynamics that can last for years after the fires occurred, and that the changes could affect deep drainage and groundwater recharge (Bellot et al., 2001; Obrist et al., 2004).

Since the processes described above occur over a range of spatial and temporal scales, the primary objective of this project was to study the soil moisture variation and dynamics in an area that bridges burned and unburned sections across a span of measurement scales. Here, two-dimensional electrical resistivity imaging (10 s of meter scale), soil moisture measurement at the ground surface (centimeter scale), and soil moisture vertical profiling (meter scale) was conducted monthly in order to determine soil moisture dynamics and their potential drivers. We address the question: How does wildfire affect soil moisture patterns and dynamics across a burn boundary and what are their potential driving factors and processes?

2. Site background and methods

2.1. Site description and background on the wildfire event

This study was conducted in Bastrop State Park which is located approximately 50 km southeast of Austin, Texas, USA (Fig. 1a). The area was devastated by a wildfire that began on September 4, 2011. The Bastrop County Complex Fire burned more than 13,760 hectares, including 96% of Bastrop State Park.

The vegetation within the 2428-hectare park is dominated by the loblolly pine (*Pinus taeda*), and is regarded as the westernmost and most arid location of the loblolly (Burns and Honkala, 1990; Retzlaff et al., 2001; Wertin et al., 2010); the region is sometimes referred to as 'The Lost Pines'. This isolated pocket of loblolly pines is within Post Oak savannah. The population of loblolly pines in the easternmost parts of Texas (east of the site) receives on average of 254 mm more rain each year than Bastrop. Thus, the loblolly pine trees in Bastrop, which receives 960 mm of rain per year, have developed adaptations to survive the drier conditions (Retzlaff et al., 2001) – including a thicker waxy coating on needles, fewer stomata, more fibrous root systems, and shorter trunks.

Soils in the studied area of the park fall into one of two classifications, according to the U.S. Department of Agriculture's Web Soil Survey. The soils are either classified as 'edge gravelly fine sandy

loam' with 3–8 percent slopes or 'Padina fine sand' with 1–12 percent slopes. The soils at the study transect (described below) were more or less uniform, as revealed by trenches, with no obvious organic layers. Grainsize analysis of soil samples collected along the study transect using a Mastersizer 3000 (Malvern Instruments Ltd., Worcestershire, U.K.) showed that the samples fall under sand or loamy sand soil texture classification; most soils from the unburned end of the transect (discussed below) were sand, and most soils from the burned end of the transect were loamy sand, but the textural differences were not very distinct. Direct gravimetric measurements of porosity ($n = 19$) showed that it ranged from 38% to 50%, typical of loamy sand, with a mean of 45%. There are no systematic spatial variations in porosity and differences between the burned and unburned ends.

Soon after the Bastrop County Complex Fire, burn severity mapping was conducted throughout the area by the Texas Parks and Wildlife Department primarily using satellite imagery and supplemented by ground validation. The classification system includes burn intensities of "Scorched," "Lightly Burned," "Moderately Burned," and "Heavily Burned," using burn severity ratings laid out by the U.S. Department of the Interior (Fig. 1b). The study transect was chosen to bridge Moderately Burned and Scorched zones (Fig. 1b) and to have minimal topographic variations. The study transect is located in the western part of the park, in a north–south swath of the park that was minimally impacted by the fire (Fig. 1b). The project began on November 12, 2012, with succeeding data collection on an approximately monthly basis and the final surveys conducted on June 6, 2013 (Fig. 2). Through the study period, and as the weather became warmer, grasses emerged across the entire transect, though they remained denser at the burned end. This period was also marked by the sprouting of saplings of both loblolly pine and post oak.

While no direct tests were done, no obvious effects of hydrophobicity were observed at the study site. This could be because the study began fourteen months after the wildfire. We also did not notice any visible remnants or effects of the fire on the surface soil materials, e.g., ash or litter or duff layers. Both observations are partly attributed to a significant storm that occurred less than a month after the fire (October 9, 2011) that had total rainfall on the order of 44 mm. The area also received several significant rainfall events in the months following the fire (Table 1), including more than 100 mm of rain over a 30-h period in January 2012. Hourly precipitation data were downloaded from a remote automated weather station (accessed through the University of Utah's MesoWest website (<http://mesowest.utah.edu>), identified as BTRT2) located approximately 10 km northwest of the field site in Bastrop State Park. Daily precipitation data are presented in Fig. 2, with survey dates indicated.

2.2. Two-dimensional electrical resistivity surveys

Electrical resistivity (ER) imaging is a frequently utilized geophysical method for mapping variation in subsurface ER which may be due to several factors such as geologic variations, soil mineralogy (e.g., clay content, metallic minerals), water content, the salinity of pore fluids, and porosity. In hydrology, for example, ER imaging has been used effectively to map soil moisture or solute dynamics (Amidu and Dunbar, 2007; Binley et al., 2002; Gasperikova et al., 2012; Miller et al., 2008; Schwartz et al., 2008; Singha and Gorelick, 2005; Zhou et al., 2001) and to investigate spatial variations associated with plant water uptake (Nijland et al., 2010) including across vegetation type boundaries (Jayawickreme et al., 2008, 2010). In this method, an electrical current is injected into the ground through two surface electrodes while other pairs of electrodes measure the electric potential field. Using these measurements, an estimate of the apparent resistivity

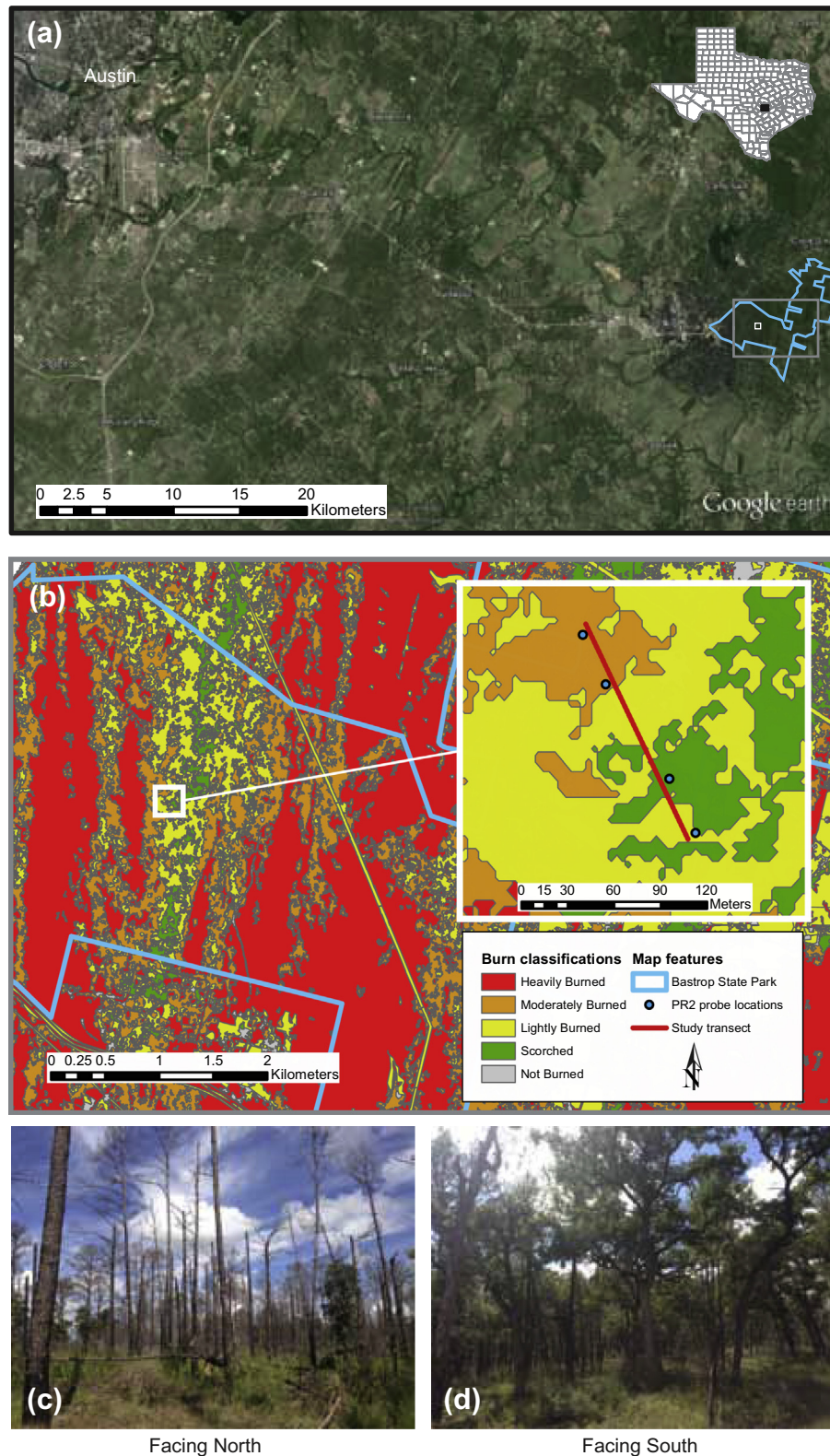


Fig. 1. Location and picture of the study site. (a) Shows the Google Earth map of the study region relative to Texas (inset); Bastrop State Park is denoted by the light-blue outline and the zoom-in map in (b) is denoted by the gray box. (b) Shows portion of a burn severity map (from the Texas Parks and Wildlife Department); the white box denotes the zoom-in area showing the study transect (red line) and some instrument locations. (c) and (d) Are pictures of the unburned and burned ends, respectively, taken from the same point at the middle of the transect. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

of the subsurface materials can be made based on Ohm's Law corrected for the geometry of the current and potential electrodes.

Two-dimensional (2D) ER surveys were conducted using the SuperSting R8 (Advanced Geosciences Inc., Austin, Texas, USA).

The study transect is oriented approximately north–south and is 165 m long, using 56 electrodes spaced 3 m apart. Both dipole–dipole and inverse Schlumberger survey configurations were conducted. Results from the two surveys were later merged into

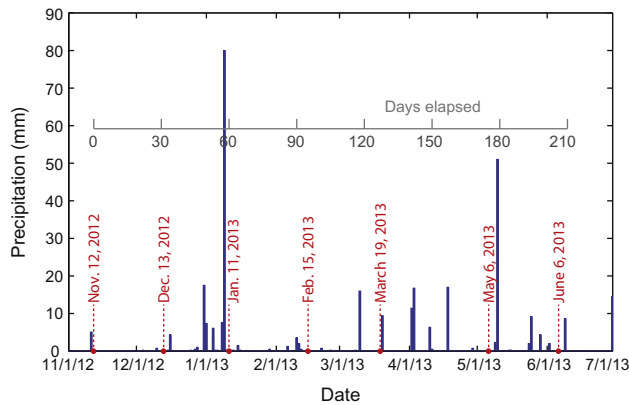


Fig. 2. Rainfall record through the study period and dates of field surveys (red). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 1

Rainfall events that occurred within a few months after the September 4, 2011 wildfire.

Dates	Rainfall (mm)	Duration (hours)
October 9, 2011	43.9	12
November 21–22, 2011	20.6	10
November 26, 2011	21.6	9
December 4–5, 2011	37.6	29
December 15–16, 2011	29.7	18
December 21–22, 2011	31.2	9
January 24–25, 2012	104.4	30
February 17–18, 2012	53.3	38
March 3–11, 2012	58.2	68
March 20, 2012	81.3	8
March 28–29, 2012	45.5	11
Total	527.3	

one file. This was done to create a more robust resistivity profile, given that dipole–dipole surveys can more effectively determine lateral ER changes, while the Schlumberger survey can better distinguish vertical variations in ER.

Surveys were completed approximately monthly, with the exception of the May survey, which was delayed due to unfavorable weather. The repeated surveys were conducted to study changes in resistivity as an indicator for changes in soil moisture, since the geology, pore water salinity, and soil texture were assumed to be constant during the study period. During the June survey, two additional transects perpendicular to the main study transect were surveyed. These transects intersect the main transect at the 15 m and 126 m points and used the same length (165 m) and electrode spacing (3 m) as the main transect.

ER measurements need to be interpreted by a fitted model. That is, an inverse model of the ER field (a tomogram) needs to be constructed that would best fit the observations of apparent ER. We implemented the inversion with RES2DINV (Loke, 2013; Loke and Barker, 1996), using the same settings for each inversion, favoring a robust inversion (Claerbout and Muir, 1973). The surface topography and exact locations of the electrodes, which were surveyed with a Trimble total station, were considered in the inversion (Tong and Yang, 1990). Data with misfit values of greater than 20% as determined by the RES2DINV software were eliminated from each set of data, and in almost all cases this was fewer than 20 data points out of ~1200; the only exception was the December 2012 survey, for which 67 data points were eliminated. The fourth inversion iteration was selected for each data set, with each having a root mean square error of less than 5%. In RES2DINV, we implemented a finite-element model for the forward model.

Three approaches are typically used for inverting and differencing time-lapse ER surveys (Miller et al., 2008): (a) inverting each dataset separately (no transfer of information from each survey) and then analyzing differences between inverted tomograms from different periods, (b) separate inversion of each surveys but using the same base or initial guess model, and (c) inverting only the ER differences of two surveys. The second approach requires a well-defined base model. The third approach works well for survey observations with low or well-characterized noise. A recent study has shown that the second and third approaches may lead to similar results in terms of resolving differences between time-lapse tomograms (Gasperikova et al., 2012). Since we were not able to constrain the noise levels a priori and since choosing a base model for a naturally dynamic environment is difficult (for example a state where the entire domain was dry would be a good base model, but that was not our case), we followed the first approach. The same finite-element mesh was used for the forward models used in each inversion; the pixelation in the resulting tomograms corresponds to the trapezoidal elements.

2.3. Soil temperature monitoring to constrain ER interpretation

ER decreases with increasing temperature (Hayley et al., 2007). Thus, we also monitored soil temperature profiles at the ends of the transect. Each profile, located 3 m to the side of the transect end points, had four HOBO TMC thermistors installed at depths of 1, 20, 40, and 60 cm connected to a HOBO U-12 data logger which logged hourly. The temperature monitoring was started a week after the first soil moisture surveys (November 19, 2012) and lasted through the study duration.

2.4. Surface soil moisture measurements

Surface volumetric soil moisture content (θ , expressed in %) was measured using a Delta-T Devices handheld ThetaProbe ML2x. The sensor has four co-axial 6 cm-long prongs which serve as waveguides that transmit an electromagnetic signal from a 100 MHz sinusoidal oscillator. The transmission of the signal through the prongs in contact with the soil allows for estimation of the soil electrical permittivity (or dielectric constant) from a voltage reading by the probe, which in turn is converted to θ via a petrophysical or calibration model (Gaskin and Miller, 1996; Topp et al., 1980). Three measurements were taken adjacent to each of the 56 electrode locations used in the ER surveys to account for local variations in θ . The measurements for each location were averaged. These measurements were repeated during each monthly ER survey.

The ThetaProbe ML2x was used with a general calibration curve for mineral soils (Delta-T Devices, 1999) instead of a soil-specific calibration. We attempted calibrating specifically using soils from the study site but the general mineral soil calibration provided by the manufacturer was deemed more reliable. Thus, all ThetaProbe and PR2 probe (described below) readings considered the suggested calibration curves for mineral soil. A general calibration produces typical errors in θ of $\pm 5\%$ (Delta-T Devices, 1999).

The discrete time-rate-of-change of θ ($\Delta\theta/\Delta t$) was calculated numerically in order to further analyze and highlight any differences between the burned and unburned ends.

2.5. Soil moisture profiling with PR2 profile probes

Soil moisture was also measured using a Delta-T Devices PR2 profile probe (Delta-T Devices, 2008). This instrument is one meter long with pairs of sensors located at 10, 20, 30, 40, 60 and 100 cm along the length of the probe and is inserted into thin-walled access tubes. The sensors work by emitting an electromagnetic

signal ~10 cm into the soil to measure its permittivity which is then converted to soil moisture (Dean et al., 1987; Whalley et al., 2004). According to Whalley et al. (2004), each of the sensor pair locations has a 100 MHz oscillator (similar to the signal transmitted by the ThetaProbe) and a detector. The detectors are sensitive to the different proportions of transmission and reflection signals, and convert them to a voltage reading which are then used to calculate θ based on a calibration curve. Using a general soil calibration curves for the PR2 probe yields typical errors in θ of $\sim\pm 6\%$. Similar to what we did with the ThetaProbe, we used manufacturer recommended coefficients for mineral soil.

Access tubes were installed adjacent to the ER survey transect on November 19, 2012, one week after the initial survey. Two tubes were placed 3 m east of the study transect at $x = 3$ m and $x = 42$ m (Fig. 1). Two additional tubes were installed 3 m west of the study transect at $x = 125$ m and $x = 159$ m. The measurements were conducted during each monthly ER survey. Three measurements were taken at each location for each survey and averaged for each location.

The total amount of water stored at each vertical profile (h in cm) was determined by calculating the average θ ($\langle\theta\rangle$) and then multiplying this by the total depth of the profile (100 cm). The $\langle\theta\rangle$ is the depth-weighted average of the point θ values; this is essentially numerical integration following the midpoint or rectangle rule. That is, a vertical support length is assigned to each measurement location which is the difference in depths between mid-points of two succeeding measurement locations. These support lengths (or weights) are 15, 10, 10, 15, 30, and 20 cm. The discrete time-rate-of-change of h ($\Delta h/\Delta t$) was calculated numerically in order to further analyze and highlight any differences between the burned and unburned ends. Furthermore, the average h for each end was calculated and the differences between the burned and unburned ends are analyzed.

2.6. Soil moisture estimates from the ER tomograms using an empirical model

A regression-based petrophysical model was developed in order to estimate θ from the ER tomograms. Since the ER inversion used trapezoidal finite-element blocks for the forward model, results are presented as block-centered data. The shallowest blocks in the tomograms are from the ground surface to 0.83 m below the ground surface, which vertically overlaps with all the PR2 readings except the deepest one (100 cm). Note, however, that the PR2 access tubes were 3 m to the side of the transect which is similar to the electrode spacing. Nevertheless, the support volume for the two datasets are similar – roughly a cubic meter for the ER tomogram block and about a half a cubic meter for the area interrogated by the PR2. Thus, the tomogram block closest to each PR2 probe was selected for development of the model. A depth-weighted average was calculated from the six PR2 measurements and then this was compared to the ER for the corresponding block through regression.

2.7. In-situ soil hydraulic conductivity measurement

During the May 6, 2013 survey, a Mini Disk Infiltrometer (Decagon Devices, Pullman, WA) was used to estimate in-situ unsaturated hydraulic conductivity along the study transect. Beginning at the unburned end of the transect ($x = 0$ m), a test was conducted adjacent to the study transect at 12 m intervals. Additional suction was imposed using the infiltrometers. This varied from 2 to 6 cm and minimized the effects of preferential flow into macropores and was used to regulate too high flow rates. Using the field data, hydraulic conductivity was calculated using Excel macros provided by the infiltrometer manufacturer and

based on Zhang (1997). The analysis requires prescribing van Genuchten parameters (Van Genuchten, 1980) for the soils and we used reported average parameters for sand soil (Carsel and Parrish, 1988) since there were no visible ash or other fire-related products nor litter layers.

3. Results

3.1. Electrical resistivity tomograms

In all the succeeding discussion, ‘depth’ pertains to depth below the ground surface. Trends in ER remained consistent throughout the study period. From the unburned end of the transect, $x = 0$ m, to approximately $x = 70$ m, and to a depth of approximately 3 m, is an area of high ER, though the higher values fluctuated throughout the study period (Fig. 3). For instance, in June 2013, the maximum ER was $\sim 13,000 \Omega\text{-m}$, while in January 2013, the maximum was $\sim 5000 \Omega\text{-m}$. These maximum ER values were observed in the unburned area of the transect. At the burned end of the transect, to a depth of about 10 m, an area of very low ER ($30 \Omega\text{-m}$ or less) was observed. In between these two areas is a region of relatively low ER (~ 100 to $\sim 300 \Omega\text{-m}$). The lowest ER area corresponds to the burned area of the transect, where all of the trees were killed, while the central area corresponds to the transition region between these two areas, where live trees are sparse.

These patterns are congruent with those from two additional ER surveys that were conducted perpendicular to the main study transect (Fig. 4) during the June 2013 survey. Here, the results at the unburned end of the transect show high ER at both the surface and in the subsurface, while the results from the burned transect show a layer of low ER material at and just below the ground surface and an overall pattern of lower resistivity (Fig. 4).

Month-to-month changes in ER are presented in Fig. 5. From November to December, the shallowest portion of the unburned end decreased in ER. The ER decrease in this area persisted through January but was then followed by an increase between January and February. This increase continued through May. Between January and February, the shallow area with ER increases was underlain by a vertically narrow area of ER reduction. The burned end of the transect showed minimal variations in ER. The deeper sections (>10 m) of the ER tomograms were mostly static.

3.2. Soil temperature and associated changes in ER

Since ER changes can also be due to temperature changes, temperature-related ER changes were analyzed. Since the lower boundary of the shallowest blocks of the ER results is at a depth of 0.83 m, only results from the deepest temperature measurements are presented and analyzed (Fig. 6). The expected percentage change in ER due to month-to-month temperature change was calculated following Eq. (2) in Hayley et al. (2007). For this analysis, we picked the temperature at noon for convenience since the surveys typically lasted from 10 am to 3 pm.

The decrease in temperature from November to January is expected to increase ER by 8.3% and 4.7% for November–December and December–January, respectively. The succeeding general warming is expected to decrease ER, from month to month. The relatively large temperature changes observed between February–March and May–June should result in an ER decrease of 12.7% and 11.6%, respectively. The expected decreases in ER between January–February and March–May are smaller; 5.4% and 2.8%, respectively. The detected ER changes between monthly surveys where dynamics were expected due to wetting/drying were larger than the ER changes that can be attributed to temperature effects (Fig. 7).

An indicator of the reliability of detecting any true differences or changes in ER is the root mean square error arising from the

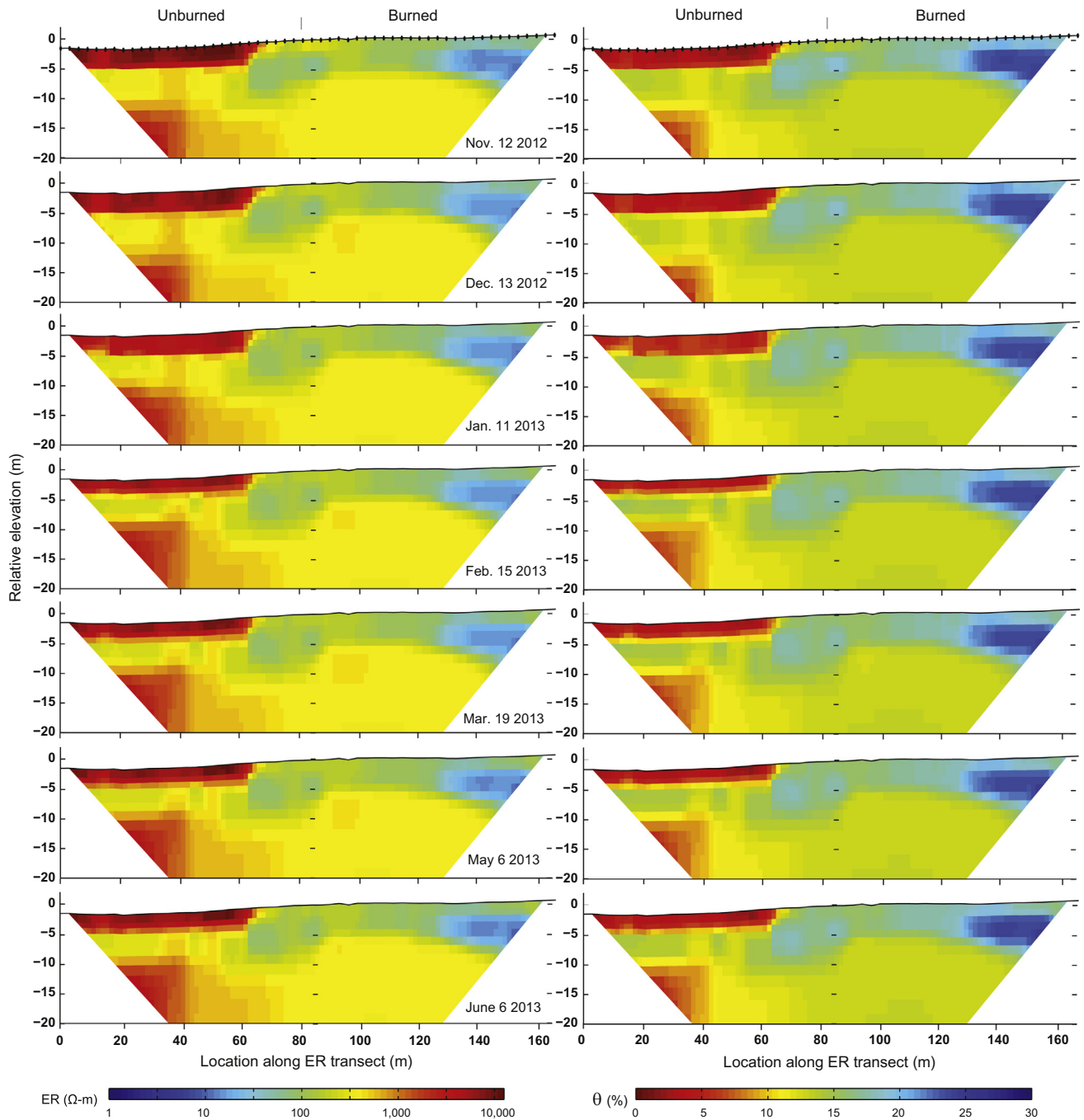


Fig. 3. Inverted electrical resistivity (ER) tomograms (left column) and calculated volumetric water content (θ) based on regression with ER (right column). The black line is the land surface with electrode locations denoted by the ellipses.

inversion. Our inverted tomograms all had a root mean square error of less than 5%. Thus, one can assume that any changes in ER, both in space and time, is within the error or detection limits of the technique. Expected ER changes due to temperature change were mostly below 10%, and this effect presumably diminishes with depth. But since some local ER changes due to observed temperature changes may be larger than 10% (Fig. 6), only ER changes larger than 10% may be interpreted reliably as a real change due to soil moisture.

3.3. Surface soil moisture from ThetaProbe measurements

In the discussion below, it is important to keep in mind that the porosity measured from surface soil samples ranged from 38% to 50% (mean of 45%). Surface θ measurements taken with the

ThetaProbe show that the unburned end of the transect was consistently drier than the burned end (Fig. 8a). While a large increase in surface θ was observed across the transect during the January survey, which was two days after a storm (Fig. 2), the contrast between the two sections of the transect persisted, even when θ decreased between surveys. The maximum θ almost always occurred around $x \sim 130$ m along the transect.

Soil moisture along the transect was relatively spatially uniform at the start of the study in November. Soil moisture generally increased throughout the transect through December and January, with the burned end showing larger increases in θ . From February onwards, the site progressively dried up. The unburned end of the transect, up to $x \sim 100$ m, was in fact almost completely dry ($\theta \sim 0\%$) at the conclusion of the study in June (Fig. 8a).

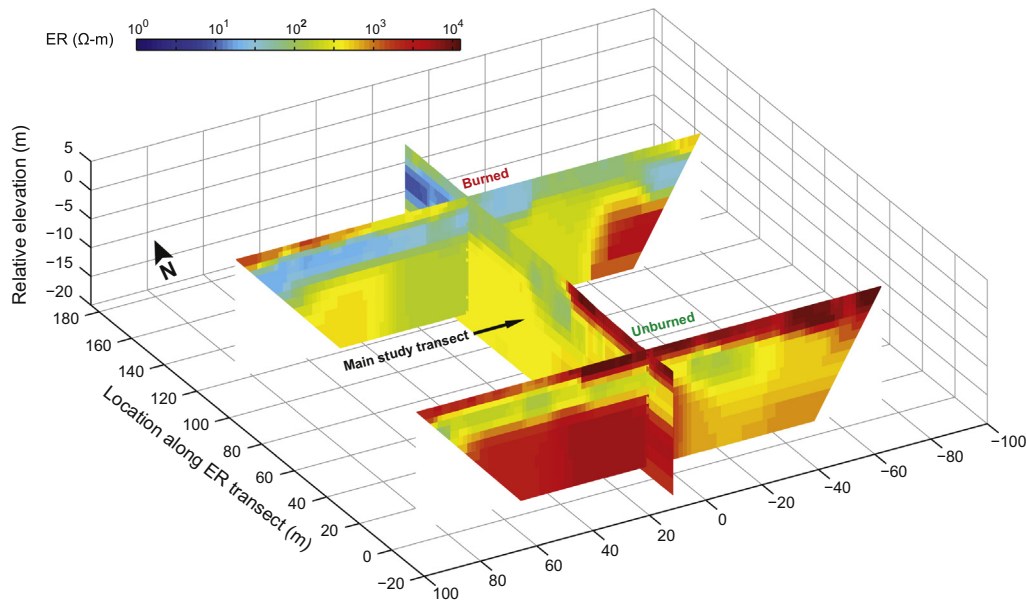


Fig. 4. Three-dimensional plot of electrical resistivity (ER) tomograms. The main study transect is crossed by two additional perpendicular transects.

3.4. Vertical soil moisture profiles from PR2 probe measurements

The temporal and spatial patterns of surface θ were further and clearly illustrated by the PR2 probe vertical soil moisture profiles for each location (Fig. 9). Soil at the unburned end of the transect located adjacent to $x = 6$ m (Fig. 9) was driest from 0 to 40 cm of depth, with somewhat higher θ in deeper soil. Soil at the second unburned location, adjacent to $x = 45$ m, was driest at 10 cm deep, with slightly wetter conditions at subsequent depths. The unburned end θ profiles showed minor local minima both at the surface at 30–40 cm depth. Over-all the patterns were similar for the two unburned locations although one was at the very end of the unburned section and the other was near the middle of the section.

Soil at the burned end of the transect, adjacent to $x = 126$ m, was wetter with depth, with soil moisture readings at the 100 cm depth consistently higher than 40%. Soil at the second probe location at the burned end of the transect, at $x = 162$ m, displayed similar patterns and was also wettest at depth. Both profiles maintained the increasing θ gradient with depth throughout the study period.

Soil moisture increased in both the burned and unburned sections beginning from November, when the study started, through January (and perhaps February) then the site began to progressively dry until the conclusion of the study. The most pronounced θ increase occurred between November and December for the unburned end, and between December and January for the burned end. During this wetting period, a local peak at a depth of 30 cm was observed. The unburned end θ recovered back to the levels of the dry November conditions by February at $x = 6$ m, and by May at $x = 45$ m. At the burned end, the soils were somewhat wetter at the conclusion of the study in June, relative to when the study started in the previous November, especially at $x = 162$ m.

3.5. Soil moisture estimates from the ER tomograms

The θ -ER regression resulted in:

$$\theta = -3.255 \cdot \ln(\text{ER}) + 32.415 \quad (1)$$

This regressed petrophysical model (Fig. 10), which has an R^2 of 0.70, was used to calculate θ from the ER tomograms. This model is more parsimonious than using Archie's Law, as typically done, since it allows direct comparison of the datasets without the uncertainty associated with the widely variable coefficients in Archie's Law (Singha and Gorelick, 2006). The empirical model leads to higher uncertainty when ER is small ($\sim 100 \Omega\text{-m}$ or smaller, Fig. 10). However, most of the ER values in the tomograms are larger than $100 \Omega\text{-m}$ (Fig. 3). On the other hand, θ becomes much less sensitive to ER when ER is very high ($>3000 \Omega\text{-m}$). Therefore, the estimated θ from the tomograms are to be considered as semi-quantitative if not qualitative.

The calculated θ showed that dry conditions were also seen at the unburned end of the transect, with increasing θ , some reaching 30%, found in soil at the burned end of the transect (Fig. 3). Except for the dry areas at the surface and directly underneath in the unburned area, the θ distribution shows a general drying trend with depth across the entire transect. This suggests the regional water table is not present within the coverage of the ER tomograms. This is reasonable since the site is 75 m above the Lower Colorado River which is 5 km south of the site; the Lower Colorado River receives groundwater baseflow and is thus a surface manifestation of the regional water table. Boreholes hand augered to 3.5 m below the ground surface at the ends of the transect were also dry, consistent with the θ from the ER tomograms.

Month-to-month θ changes based on the ER tomograms are shown in Fig. 5. The most pronounced change was the increased θ from December–January to January–February. This correlates with the heavy rainfall that occurred two days before the January survey and previous expected infiltration events subsequent to the large rain event (Fig. 2).

In extracting and examining the shallowest ER block data from the tomograms, the same trend of high ER at the unburned end of the transect and low ER at the burned end of the transect is apparent (Fig. 8c). When these data are converted to θ using Eq. (1), the lower θ at the unburned end of the transect and higher θ at the burned end are more obvious (Fig. 8b). This is similar to surface θ measured with the ThetaProbe (Fig. 8a), though the correlation is not exact since the ER data is for a $0.7 \text{ m} \times 0.83 \text{ m}$ block, while

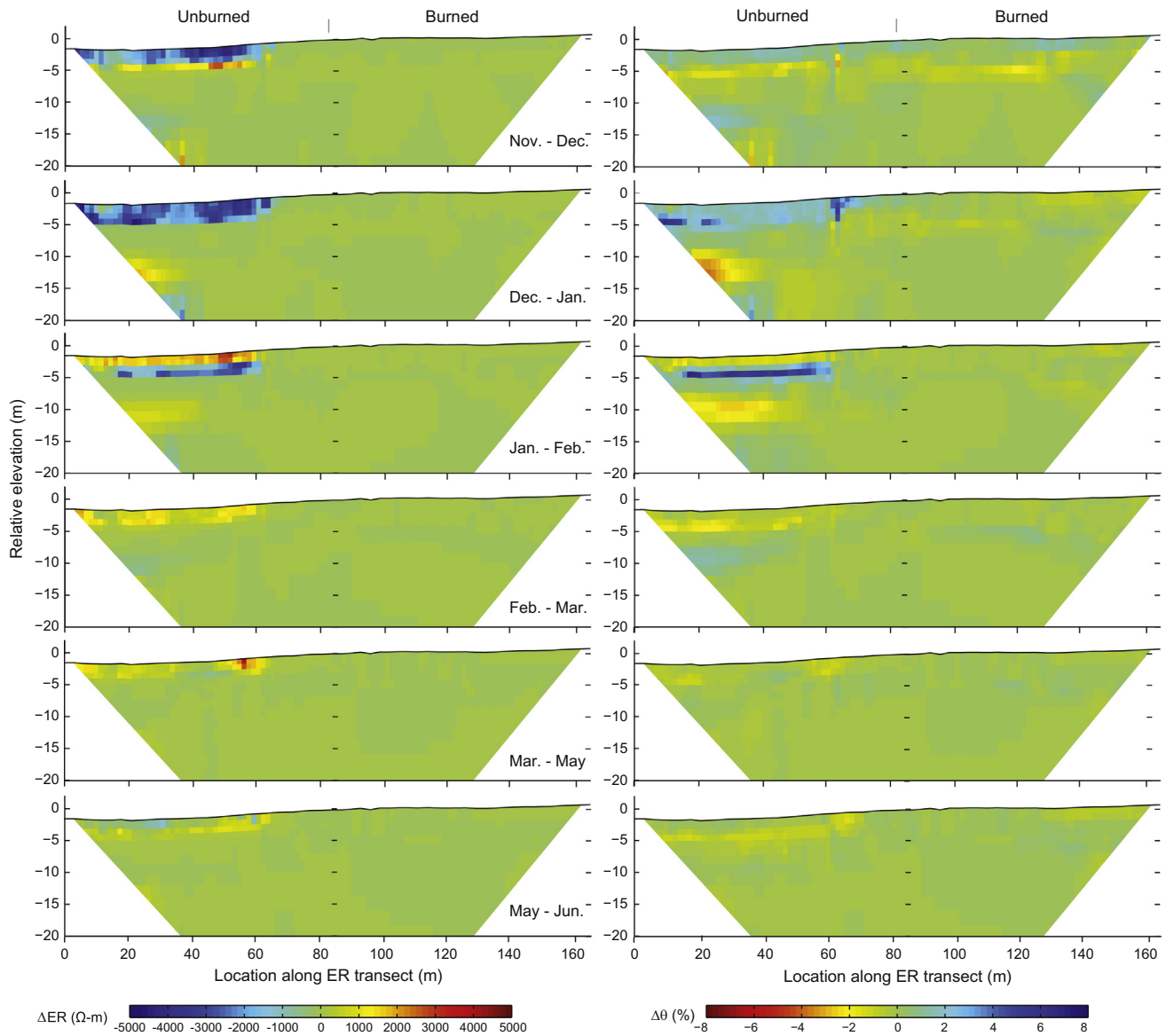


Fig. 5. Absolute differences in electrical resistivity (ΔER) between surveys (left column) and corresponding differences in volumetric water content ($\Delta \theta$) (right column).

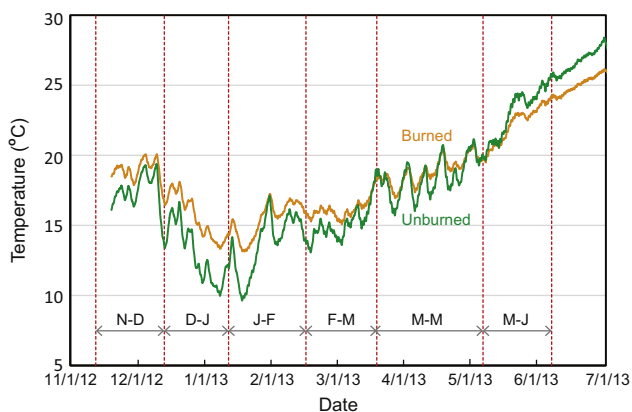


Fig. 6. Soil temperature at 60 cm depth from the burned (orange) and unburned (green) sections of the transect; dashed red lines denote when electrical resistivity (ER) surveys were conducted. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

the ThetaProbe measurement volume is from the top 6 cm of the soil.

3.6. Comparison of soil moisture measurements

The unburned part was consistently drier based on all three methods used for estimating θ (ThetaProbe, PR2, and ER). The temporal changes in θ were also mostly consistent. The surface θ increased from November to January (Fig. 8a). This increase was also detected at the topmost and shallowest ER measurements (Figs. 3, 5 and 8b). Likewise, this was also observed in the PR2 θ profiles (Fig. 9). All measurements then indicated a progressive drying of the site after peak θ in February. However, the ER images suggested less pronounced changes in θ . This is expected since ER imaging not only detects and integrates signals over a larger area/volume (typically considered as half the electrode spacing in the shallow portions of the tomogram; so in this case the resolution is 1.5 m), the inversion process also makes sharp boundaries more diffuse. Note however that we implemented a robust inversion which preserves interfaces better.

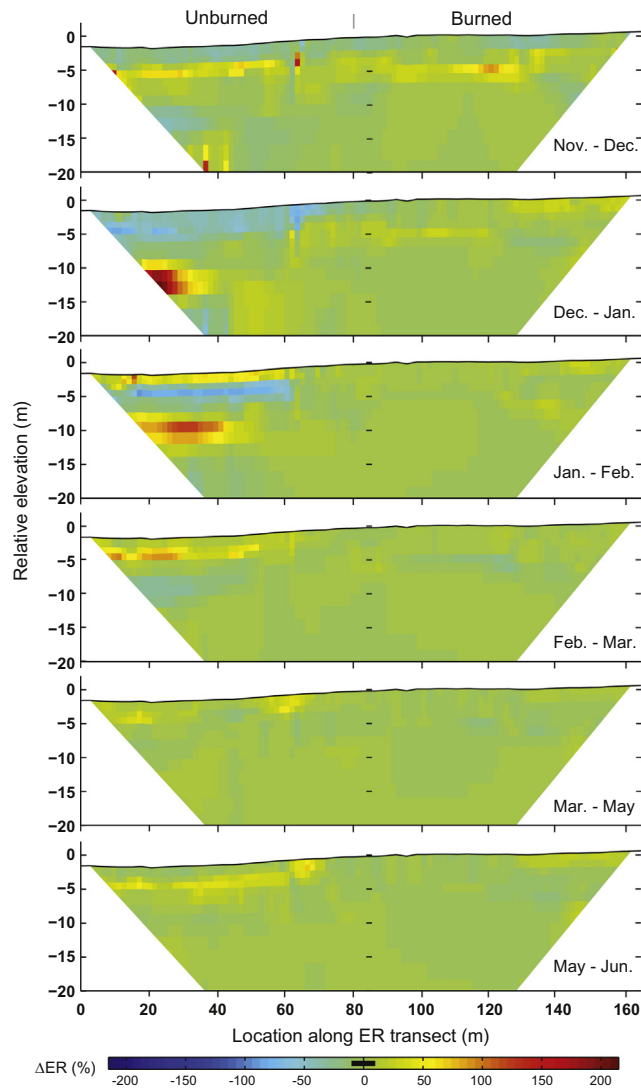


Fig. 7. Observed percent changes in ER between surveys. The black bar from -10% to 10% indicates the range where changes within are either due to the limited sensitivity of the survey and inversion or changes can be solely due to temperature variation (see Fig. 6).

A relatively rainy period in early January which culminated in the largest rainfall observed during the study period (Fig. 2) resulted in different responses between the two sections, although both ends became wetter suggesting infiltration. This infiltration event is particularly evident in the PR2 (Fig. 9) and ER profiles (Figs. 3 and 4) which suggest moisture increases up to depths of larger than 1 m. Despite the over-all wetting of the study transect, there were clear differences between the unburned and burned sections. The burned sections showed a more or less uniform increase in θ with depth (Fig. 9). On the other hand, the burned end appears to show a wetting front going down the profile. In particular, this is suggested by a peak in θ at intermediate depths (Fig. 9). Except for during January, the θ in the unburned section mostly stayed below 10% , both at the surface (Fig. 8) and down to 100 cm (Fig. 9), whereas θ was above 10% in the burned area and was larger than 15% from 50 to 100 cm (Fig. 9). At $x = 162$ m, the θ at 100 cm depth remained at the maximum of $\sim 35\%$ for four months (February–May). This suggests near-saturated or even saturated conditions.

The temporal dynamics of θ was further analyzed by taking the time-rate-of-change of the observations from the surface

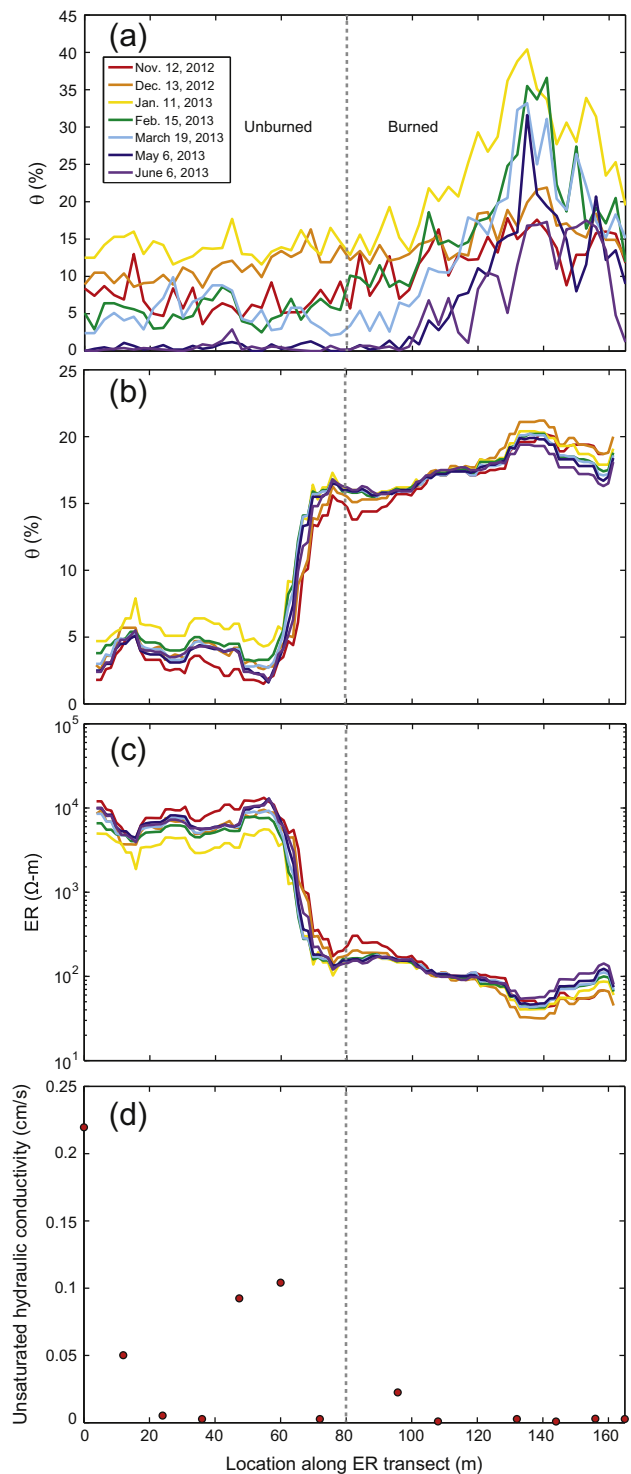


Fig. 8. Various parameters at or near the soil surface across the transect from different measurement periods: (a) soil moisture (θ) at the surface measured with the ThetaProbe, (b) calculated θ based on (c) electrical resistivity (ER) from the topmost block of the inverted tomograms (depth = 0 – 0.83 m from the ground surface), and (d) unsaturated hydraulic conductivity estimated with MiniDisk infiltrometer measurements.

measurements (Fig. 8a) and vertical profiles (Fig. 9), $\Delta\theta/\Delta t$. Furthermore, we calculated the total equivalent depth of water (h , in cm) stored within the vertical profiles through their numerical integration. From this, we also calculated the time-rate-of-change $\Delta h/\Delta t$. This analysis further revealed the differences between the two ends.

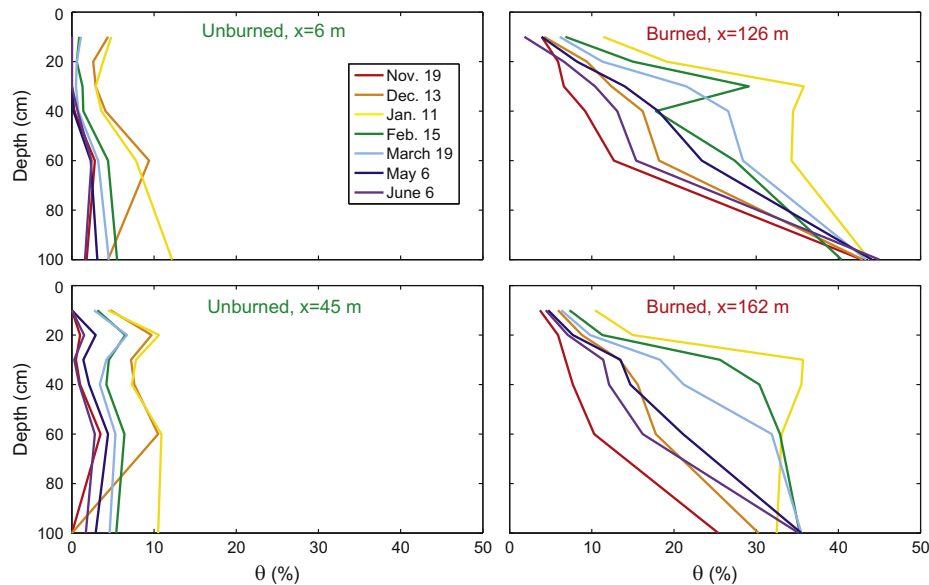


Fig. 9. Vertical soil moisture (θ) profiles at four locations (x) within the transect from different measurement periods. Measurements were done with a PR2 probe.

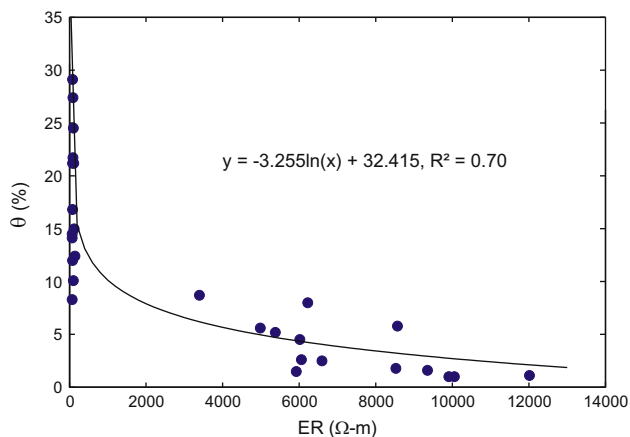


Fig. 10. Regression model for soil moisture (θ) based on electrical resistivity (ER). The comparisons are based on PR2 probe measurements and their nearest ER tomogram block center.

The surface changes in θ generally occurred at a higher rate in the burned end (Fig. 11a), particularly after the January rainstorms (Fig. 2) when wetting occurred, but also during the late spring to summer (March–June). The unburned end surface θ was more static, with smaller temporal rates of change, which is largely due to its drier state. Accounting for h likewise further showed the generally much drier state of the unburned end (Fig. 11b), which translates to lower rates in $\Delta h/\Delta t$. There is roughly 2–3 times more water in the burned end (compare Fig. 11b and c). The analysis also showed that a substantial amount of the water input from the January rainstorms (Fig. 2) infiltrated in the burned end (Fig. 11c), but a much smaller portion of this was reflected in the unburned end moisture storage (Fig. 11b). The unburned end dried down quickly whereas the unburned end appears to still be drying down at the conclusion of the study.

The contrasting soil moisture regime is most obvious in the differences of the amount of water stored (h_{diff} = burned end average h minus the unburned end average h) and how this evolved through time (Fig. 11b). The burned end clearly always had more water stored (7.2 cm at the start of the study, and 8.2 cm at the end of the study) and this difference peaked at 12.4 cm after the

storms in January; this amount represents 13% of mean annual precipitation.

3.7. In-situ soil hydraulic conductivity

Data collected using the Mini Disk Infiltrator were used to calculate unsaturated hydraulic conductivity (K_u) for fifteen points along the study transect (Fig. 8d). The variability in K_u not only reflects differences in soil texture but also differences due to the moisture state, i.e., the dependence of K_u on θ . The reported values reflect the in-situ conditions when the infiltrator tests were done (May 2013). The average K_u for the study area is 0.03 cm/s (~ 27 m/d), which is within the range for saturated hydraulic conductivity of well-sorted sand. Regardless of and excluding a high value for K_u at $x = 0$ m, the unburned section tended to have higher K_u . The average K_u at the unburned and burned ends were 0.068 cm/s and 0.006 cm/s, respectively.

4. Discussion

4.1. Potential causes of similarities and differences across the transect

Why are the two ends hydrologically different? The obvious explanation is this is due to burning's effects on the soils. There are two key observations relating to θ that differ between the unburned and burned ends: (1) the unburned end θ increased less but uniformly relative to the burned end during the same times (i.e., responding to the same atmospheric and surface forcing), and (2) the burned end maintained high θ at depth and, except for during infiltration after the wet January, maintained a gradient of increasing θ with depth. Both phenomena can be explained by the death of trees in the burned end, as can be seen in Fig. 1c. We postulate that the loss of trees and canopy cover led to less canopy interception and transpiration in the burned end. Both effects would lead to the same result of increased θ in the burned end. Rain throughfall has been shown to increase with decreasing stand density for loblolly pine (Stogsdill et al., 1989) which is also accompanied by soil moisture increase (Stogsdill et al., 1992). In shrublands in Spain, burned areas had 88% throughfall, while unburned areas had 58% throughfall (Soto and Diaz-Fierros, 1997); this led to higher soil moisture in the burned areas. Silva

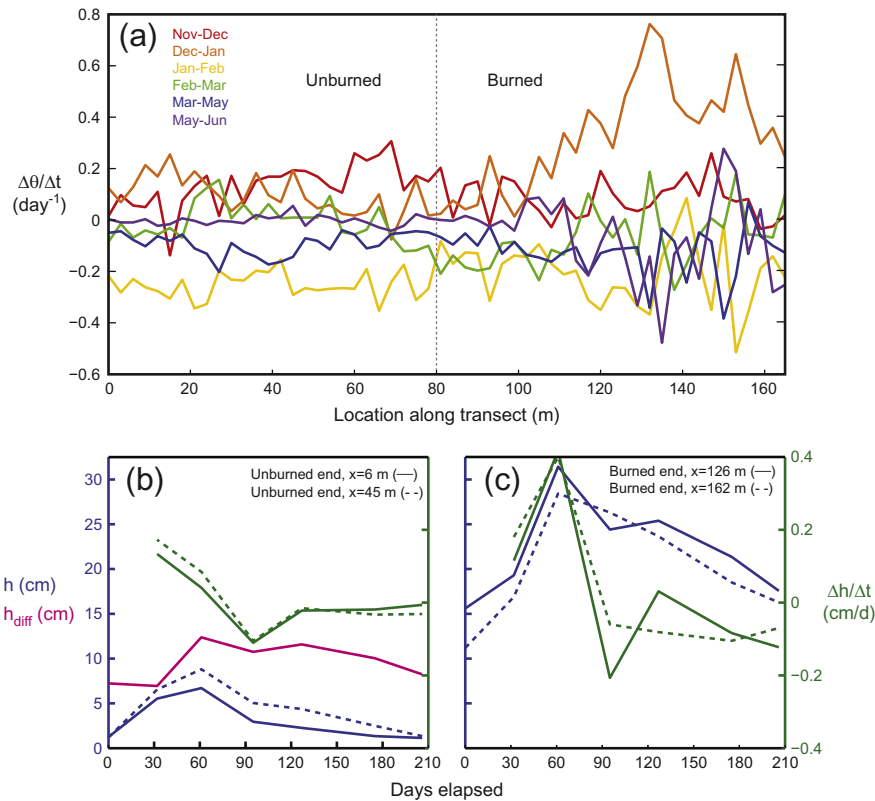


Fig. 11. (a) Rate of change in soil moisture ($\Delta\theta/\Delta t$), taken as the difference between surveys) based on the approximately monthly surface soil moisture measurements shown in Fig. 8a. (b) Shows total water stored (h in cm; left blue axes) and the difference between the burned and unburned ends (h_{diff}), and (c) its rate of change ($\Delta h/\Delta t$, right green axes) for the 100-cm deep vertical soil moisture profiles shown in Fig. 9 at the unburned and burned ends, respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

et al. (2006) observed similar soil moisture dynamics of wetter burned areas due to experimental fires. While we did not measure throughfall or quantified the canopy cover, e.g., leaf area index, the difference between the two ends is visibly obvious (Fig. 1c and d).

Actual evapotranspiration rates from loblolly stands in East Texas, which represents a wetter climate relative to our study site, range from 70 to 100 cm/year. The mean annual rainfall at the study site of 96 cm/year is within this range. The annual potential evapotranspiration in the Austin, Texas area is 146 cm (<http://texaset.tamu.edu/pet.php>) whereas the annual actual evapotranspiration in the study area (Bastrop county) is 71–80 cm (Sanford and Selnick, 2013), which is 74–83% of the annual rainfall. This strong water deficit suggests that the trees in Bastrop are likely to consume all available water. And at our site, where the regional water table is well below the range of any taproots, the water used is expected to be exclusively soil moisture from the vadose zone. The roots of loblolly pine consists of a taproot that is normally 1 m deep but this may go deeper at times (Retzlaff et al., 2001). However, the extensive lateral root network is typically confined to the upper 40 cm with over 70% of all root biomass in the upper 0–20 cm of soil. In some studies, no lateral roots are typically observed beyond 53 cm (Burns and Honkala, 1990). Thus, when water is available, loblolly pine tends to use the water in the upper 40 cm (Retzlaff et al., 2001). In fact, even under irrigated conditions, loblolly pine was shown to consume more than 90% of its water from the upper 75 cm of soil, more than 79% from the upper 35 cm, and with less than 10% of the water used from deeper than 1 m (Gonzalez-Benecke and Martin, 2010). Under dry conditions, loblolly pine can quickly use any available water (Retzlaff et al., 2001). The patterns we observed are consistent with expected patterns based on the root distribution and root water uptake

dynamics of loblolly pine. The upper 40 cm at the unburned site remained mostly dry (Fig. 9) – this coincides with where loblolly pine typically gets water from and is where its lateral roots are concentrated. In fact, there tended to be a minimum in θ between 30 and 40 cm depth at the unburned ends (Fig. 9); this suggests a moisture sink at this interval which is consistent with root water uptake by loblolly pine.

The relatively high potential evapotranspiration rates should also be apparent as primarily evaporation in the burned area, which has been stripped of its canopy cover (Fig. 1c). The persistent and strong gradient in the θ profiles of the burned end (Fig. 9) indicates moisture flow towards the surface. This along with the higher drying-down rates (Fig. 11) are consistent with evaporation. The surface is always the driest in these profiles. This pattern persisted even when there was infiltration, wetting, and redistribution in the subsurface.

The explanations we propose cannot be completely proven as canopy interception and actual evapotranspiration were not measured across the transect. However, our conceptual interpretation is consistent with the previous studies described above. A recent synthetic modeling study driven by measurements (Ebel, 2013) also presents a physically-based picture consistent with our interpretation. It was shown through numerical simulations that decreases in canopy interception and evapotranspiration in burned areas led to increased soil moisture and groundwater recharge (i.e., deep moisture drainage).

Hydrophobicity is also a potential explanation for the differences between burned and unburned ends. However, no obvious evidence for hydrophobicity were observed, i.e., no water beads form when water was poured on the surfaces. Moreover, several storms (Table 1) after the fire could have caused erosion and

removal of any easily erodible surficial materials such as ash and other hydrophobic materials. In fact, there were no obvious ash layers and there were no obvious differences in remnant litter. By the time the study started, both ends had mostly bare sand soil, except for some grasses and mosses in some wetter spots. The study also started fourteen months after the fire which is past the normal period when soils experience post-fire hydrophobicity (DeBano, 2000; Neary et al., 2005). Therefore, differential infiltration due to hydrophobicity is unlikely reason.

Another potential explanation for the differences is that they are due to different soil textures. The sand soil in the unburned end is slightly coarser than the loamy sand in the burned end. This texture difference is also evident in the soil unsaturated hydraulic conductivity (Fig. 8d). Thus, any water inputs should be observed sooner if not more prominently in the unburned end. This is not the case based on the PR2 θ and ER profiles (Fig. 9). Not only is there more moisture in the burned end but the change in surface θ (and its rate of change) is also larger. The coarser sand at the unburned end is also more likely to exhibit a sharper wetting front which was not observed. Therefore, it is unlikely that the observed differences are due primarily or solely due to differences in texture. In fact, it is paradoxical that loamy sand is behaving as more hydraulically conductive than sand. Perhaps part of this behavior may be due to dead and burnt roots in the burned end acting as macropores, e.g., Noguchi et al. (1999), Nyman et al. (2010, 2014), Beven and Germann (2013); or perhaps due to the occurrence of preferential or fingered flow such as observed in other burned soils (Stoof et al., 2014). Live roots in the unburned end would also serve as macropores. However, these roots would also uptake the moisture unlike dead roots. Loblolly pine has extensive lateral roots in the upper 40 cm of soils with most of its root biomass, including the tap root, typically concentrated in the upper meter (Retzlaff et al., 2001). Other factors must be at play in order to broadly explain our results.

4.2. Implications on soil ecohydrology and water resources

Our findings have implications on ecological processes particularly for vegetation recovery. The results indicate that burned areas will be wetter and store more water mainly due to less canopy interception of rainfall and elimination of tree transpiration due to their death. Thus, this makes more water available for new vegetation. It is noteworthy that grasses, shrubs, and tree saplings have begun to grow in the area. As these plants grow, they will have the benefit of initially having more water in the burned areas.

The wetter burned areas may act as enhanced groundwater recharge zones. In semi-arid areas, soil water that makes it past the root zone is typically assumed to eventually reach the regional water table and is thus considered as recharge to the aquifer. The burned areas no longer have a well-established root zone, at least in the months following the fire. This allows for water that would normally be sequestered by plants and transpired back into the atmosphere to infiltrate deeper into the soil and potentially recharge the underlying aquifers. The Bastrop County Complex Fire burned more than 13,760 hectares. These burned areas, in the short-term may serve to enhance recharge, particularly due to large rain events, since ordinarily most water that enters the soil is consumed by the water-stressed trees. This would be a temporary effect until vegetation recovers which would exploit the newly available water.

5. Summary and conclusions

We monitored soil moisture across a 165-m long transect that crossed a wildfire boundary between burned and unburned

sections in a loblolly pine forest using a variety of electrical and dielectric methods including: surface measurements using a handheld probe, vertical profiles using profiling probes, and 2D electrical resistivity imaging. Trees remained alive in the unburned end whereas they are mostly dead in the burned end. Seven approximately monthly measurements were started fourteen months after the fire. The burned end was much wetter throughout the study period and had a more pronounced infiltration and wetting response due to rainfall. The burned end also maintained a downward wetting gradient in moisture consistent with evaporation at the surface. On the other hand, the unburned end, which was much drier than the burned end, had soil moisture profiles close to vertically uniform down to a depth of 1 m, with minor local moisture minima both at the surface and at about 30–40 cm. This is consistent with root water uptake by the loblolly pine which typically sources its moisture from the upper 40 cm where its lateral roots are concentrated. The different responses to rainfall also indicate differences in water inputs. This may be explained by the loss of canopy cover and therefore more rain throughfall in the burned end, whereas canopy interception might have occurred in the unburned end. Thus, the burning and death of trees has likely led to more moisture inputs (less canopy interception) and less outputs (little to no root water uptake by trees), resulting in increased water storage in the burned end. This additional moisture will be helpful for plants reoccupying the burned areas and may also lead to a temporary increase in regional groundwater recharge.

Acknowledgements

We thank Greg Creacy and the staff of the Texas Parks and Wildlife Department, especially those at Bastrop State Park, for logistical support and access, and for guidance with establishing the field site. Michael Young and John Sharp commented on earlier versions of this manuscript. Kevin Befus provided assistance in the analysis of the data using Matlab. We thank Kevin Befus, Alyse Briody, Wen Deng, Raquel Flinker, Brad Gooch, Matt Kaufman, Wendy Robertson, Ram Sanchez, Stacy Slater, Kris Voorhees, Lichun Wang, Peter Zamora, and Lizhi Zheng for assistance with field work. This research was partly supported by the Geology Foundation at the University of Texas at Austin.

References

- Amidu, S.A., Dunbar, J.A., 2007. Geoelectric studies of seasonal wetting and drying of a Texas vertisol. *Vadose Zone J.*, 511–523.
- Bellot, J., Benet, A., Sanchez, J.R., Chirino, E., 2001. Likely effects of land use changes on the runoff and aquifer recharge in a semiarid landscape using a hydrological model. *Landscape Urban Plann.* 55 (1), 41–53.
- Bento-Goncalves, A., Vieira, A., Ubeda, X., Martin, D., 2012. Fire and soils: key concepts and recent advances. *Geoderma* 191, 3–13.
- Beven, K., Germann, P., 2013. Macropores and water flow in soils revisited. *Water Resour. Res.*, 3071–3092.
- Binley, A., Winship, P., West, L.J., Pokar, M., Middleton, R., 2002. Seasonal variation of moisture content in unsaturated sandstone inferred from borehole radar and resistivity profiles. *J. Hydrol.* 267 (3–4), 160–172.
- Bowman, D.M.J.S. et al., 2009. Fire in the Earth system. *Science*, 481–484.
- Burns, R.M., Honkala, B.H., 1990. *Silvics of North America*, vol. 1: conifers. U.S. Department of Agriculture, Forest Service, Washington, D.C.
- Carsel, R.F., Parrish, R.S., 1988. Developing joint probability-distributions of soil-water retention characteristics. *Water Resour. Res.* 24 (5), 755–769.
- Certini, G., 2005. Effects of fire on properties of forest soils: a review. *Oecologia*, 1–10.
- Claerbout, J.F., Muir, F., 1973. Robust modeling with erratic data. *Geophysics*, 826–844.
- Dean, T.J., Bell, J.P., Baty, A.J.B., 1987. Soil-moisture measurement by an improved capacitance technique. 1. Sensor design and performance. *J. Hydrol.*, 67–78.
- DeBano, L.F., 2000. The role of fire and soil heating on water repellency in wildland environments: a review. *J. Hydrol.*, 195–206.
- Delta-T Devices, 1999. *Theta Probe Soil Moisture Sensor User Manual*. Delta-T Devices Ltd., Cambridge UK.
- Delta-T Devices, 2008. *User Manual for the Profile Probe type PR2*. Delta-T Devices Ltd., Cambridge UK.

- Doerr, S.H., Shakesby, R.A., Walsh, R.P.D., 2000. Soil water repellency: its causes, characteristics and hydro-geomorphological significance. *Earth – Sci. Rev.*, 33–65.
- Ebel, B.A., 2013. Simulated unsaturated flow processes after wildfire and interactions with slope aspect. *Water Resour. Res.* 49 (12), 8090–8107.
- Ebel, B.A., Moody, J.A., 2013. Rethinking infiltration in wildfire-affected soils. *Hydrol. Proc.*, 1510–1514.
- Gaskin, G.J., Miller, J.D., 1996. Measurement of soil water content using a simplified impedance measuring technique. *J. Agr. Eng. Res.*, 153–159.
- Gasperikova, E. et al., 2012. Long-term electrical resistivity monitoring of recharge-induced contaminant plume behavior. *J. Contam. Hydrol.* 142, 33–49.
- Gonzalez-Benecke, C.A., Martin, T.A., 2010. Water availability and genetic effects on water relations of loblolly pine (*Pinus taeda*) stands. *Tree Physiol.* 30 (3), 376–392.
- Gonzalez-Perez, J.A., Gonzalez-Vila, F.J., Almendros, G., Knicker, H., 2004. The effect of fire on soil organic matter – a review. *Environ. Int.*, 855–870.
- Hayley, K., Bentley, L.R., Gharibi, M., Nightingale, M., 2007. Low temperature dependence of electrical resistivity: implications for near surface geophysical monitoring. *Geophys. Res. Lett.*
- Jayawickreme, D.H., Van Dam, R.L., Hyndman, D.W., 2008. Subsurface imaging of vegetation, climate, and root-zone moisture interactions. *Geophys. Res. Lett.*
- Jayawickreme, D.H., Van Dam, R.L., Hyndman, D.W., 2010. Hydrological consequences of land-cover change: quantifying the influence of plants on soil moisture with time-lapse electrical resistivity. *Geophysics*, WA43–WA50.
- Kinner, D.A., Moody, J.A., 2010. Spatial variability of steady-state infiltration into a two-layer soil system on burned hillslopes. *J. Hydrol.*, 322–332.
- Letey, J., 2001. Causes and consequences of fire-induced soil water repellency. *Hydrol. Proc.*, 2867–2875.
- Loke, M.H., 2013. RES2DINV. Geotomo Software, Penang, Malaysia.
- Loke, M.H., Barker, R.D., 1996. Rapid least-squares inversion of apparent resistivity pseudosections by a quasi-Newton method. *Geophys. Prospect.*, 131–152.
- Miller, C.R., Routh, P.S., Brosten, T.R., McNamara, J.P., 2008. Application of time-lapse ERT imaging to watershed characterization. *Geophysics* 73 (3), G7–G17.
- Moody, J.A., Martin, D.A., 2001a. Initial hydrologic and geomorphic response following a wildfire in the Colorado Front Range. *Earth Surf. Proc. Land.*, 1049–1070.
- Moody, J.A., Martin, D.A., 2001b. Post-fire, rainfall intensity-peak discharge relations for three mountainous watersheds in the western USA. *Hydrol. Proc.*, 2981–2993.
- Moody, J.A., Martin, D.A., 2009. Synthesis of sediment yields after wildland fire in different rainfall regimes in the western United States. *Int. J. Wildland Fire*, 96–115.
- Moody, J.A., Kinner, D.A., Ubeda, X., 2009. Linking hydraulic properties of fire-affected soils to infiltration and water repellency. *J. Hydrol.*, 291–303.
- Moody, J.A., Shakesby, R.A., Robichaud, P.R., Cannon, S.H., Martin, D.A., 2013. Current research issues related to post-wildfire runoff and erosion processes. *Earth – Sci. Rev.*, 10–37.
- Neary, D.G., Ryan, K.C., DeBano, L.F., 2005. *Wildland Fire in Ecosystems: Effects of Fire on Soils and Water*. USDA, Ogden UT.
- Nijland, W., van der Meijde, M., Addink, E.A., de Jong, S.M., 2010. Detection of soil moisture and vegetation water abstraction in a Mediterranean natural area using electrical resistivity tomography. *Catena*, 209–216.
- Noguchi, S., Tsuboyama, Y., Sidle, R.C., Hosoda, I., 1999. Morphological characteristics of macropores and the distribution of preferential flow pathways in a forested slope segment. *Soil Sci. Soc. Am. J.* 63 (5), 1413–1423.
- Nolan, R.H., Lane, P.N.J., Benyon, R.G., Bradstock, R.A., Mitchell, P.J., in press. Changes in evapotranspiration following wildfire in resprouting eucalypt forests, *Ecohydrology*. <http://dx.doi.org/10.1002/eco.1463>.
- Nyman, P., Sheridan, G., Lane, P.N.J., 2010. Synergistic effects of water repellency and macropore flow on the hydraulic conductivity of a burned forest soil, south-east Australia. *Hydrol. Proc.*, 2871–2887.
- Nyman, P., Sheridan, G.J., Smith, H.G., Lane, P.N.J., 2014. Modeling the effects of surface storage, macropore flow and water repellency on infiltration after wildfire. *J. Hydrol.*, 301–313.
- Obrist, D., Yakir, D., Arnone, J.A., 2004. Temporal and spatial patterns of soil water following wildfire-induced changes in plant communities in the Great Basin in Nevada, USA. *Plant Soil*, 1–12.
- Pechony, O., Shindell, D.T., 2010. Driving forces of global wildfires over the past millennium and the forthcoming century. *Proc. Natl. Acad. Sci. USA*, 19167–19170.
- Retzlaff, W.A., Blaisdell, G.K., Topa, M.A., 2001. Seasonal changes in water source of four families of loblolly pine (*Pinus taeda* L.). *Trees-Struct. Funct.*, 154–162.
- Robichaud, P.R., 2000. Fire effects on infiltration rates after prescribed fire in Northern Rocky Mountain forests, USA. *J. Hydrol.*, 220–229.
- Sanford, W.E., Selnick, D.L., 2013. Estimation of evapotranspiration across the conterminous United States using a regression with climate and land-cover data. *J. Am. Water Resour. As.*, 217–230.
- Schwartz, B.F., Schreiber, M.E., Yan, T.T., 2008. Quantifying field-scale soil moisture using electrical resistivity imaging. *J. Hydrol.*, 234–246.
- Silva, J.S., Rego, F.C., Mazzoleni, S., 2006. Soil water dynamics after fire in a Portuguese shrubland. *Int. J. Wildland Fire*, 99–111.
- Singha, K., Gorelick, S.M., 2005. Saline tracer visualized with three-dimensional electrical resistivity tomography: field-scale spatial moment analysis. *Water Resour. Res.* 41 (5).
- Singha, K., Gorelick, S.M., 2006. Effects of spatially variable resolution on field-scale estimates of tracer concentration from electrical inversions using Archie's law. *Geophysics* 71 (3), G83–G91.
- Soto, B., Diaz-Fierros, F., 1997. Soil water balance as affected by throughfall in gorse (*Ulex europaeus*, L.) shrubland after burning. *J. Hydrol.*, 218–231.
- Stogsdill, W.R., Wittwer, R.F., Hennessey, T.C., Dougherty, P.M., 1989. Relationship between throughfall and stand density in a *Pinus taeda* plantation. *Forest Ecol. Manag.*, 105–113.
- Stogsdill, W.R., Wittwer, R.F., Hennessey, T.C., Dougherty, P.M., 1992. Water use in thinned loblolly pine plantations. *Forest Ecol. Manag.* 50 (3–4), 233–245.
- Stoof, C.R., Slingerland, E.C., Mol, W., van den Berg, J., Vermeulen, P.J., Ferreira, A.J.D., Ritsema, C.J., Parlange, J.Y., Steenhuis, T.S., 2014. Preferential flow as a potential mechanism for fire-induced increase in streamflow. *Water Resour. Res.*, 1840–1845.
- Tong, L.T., Yang, C.H., 1990. Incorporation of topography into 2-dimensional resistivity inversion. *Geophysics*, 354–361.
- Van Genuchten, M.T., 1980. A closed-form equation for predicting the hydraulic conductivity of unsaturated soils. *Soil Sci. Soc. Am. J.* 44 (5), 892–898.
- Topp, G.C., Davis, J.L., Annan, A.P., 1980. Electromagnetic determination of soil-water content – measurements in coaxial transmission lines. *Water Resour. Res.*, 574–582.
- Varela, M.E., Benito, E., Keizer, J.J., 2010. Effects of wildfire and laboratory heating on soil aggregate stability of pine forests in Galicia: the role of lithology, soil organic matter content and water repellency. *Catena*, 127–134.
- Werten, T.M., McGuire, M.A., Teskey, R.O., 2010. The influence of elevated temperature, elevated atmospheric CO₂ concentration and water stress on net photosynthesis of loblolly pine (*Pinus taeda* L.) at northern, central and southern sites in its native range. *Glob. Change Biol.*, 2089–2103.
- Westerling, A.L., Hidalgo, H.G., Cayan, D.R., Swetnam, T.W., 2006. Warming and earlier spring increase western US forest wildfire activity. *Science*, 940–943.
- Whalley, W.R., Cope, R.E., Nicholl, C.J., Whitmore, A.P., 2004. In-field calibration of a dielectric soil moisture meter designed for use in an access tube. *Soil Use Manage.*, 203–206.
- Zavala, L.M., Granged, A.J.P., Jordan, A., Barcenas-Moreno, G. (Eds.), 2010. Effect of burning temperature on water repellency and aggregate stability in forest soils under laboratory conditions. *Geoderma* 158, 366–374.
- Zhang, R.D., 1997. Determination of soil sorptivity and hydraulic conductivity from the disk infiltrometer. *Soil Sci. Soc. Am. J.* 61 (4), 1024–1030.
- Zhou, Q.Y., Shimada, J., Sato, A., 2001. Three-dimensional spatial and temporal monitoring of soil water content using electrical resistivity tomography. *Water Resour. Res.* 37 (2), 273–285.